

AI in the Public Sector

Experience and challenges from developing countries

Methodology

Qualitative Multi Method Approach with Two Stages:

- **Narrative Literature Review:** what is known about AI in the public sector of developing countries, and how this compares with trends identified in developed economies.
- **Two Case Studies of Social Protection institutions in Latin America:** The Superintendency of Social Security (SUSESO) in Chile and the National Planning Department (DNP) in Colombia.

Theoretical Framework: the Four AI Challenges model developed by Wirtz et al. (2019), which is focused on public organizations.

Developed and Developing Countries

- Developed economies tend to enjoy more mature technological infrastructure and well prepared experts to deal with AI development.
- In developing countries, especially within the public sector, the lack of AI specialists creates a dependency on external data science providers. In some cases, this can lead to seeking support from experts abroad.
- Regarding AI regulations, research suggests that developed countries focus on compliance with existing legal frameworks, which could even work as a boundary when too restrictive.
- In the case of developing countries, weak or even absent regulatory systems are observed. This makes these systems more vulnerable to privacy and ethical issues.

Insights from Case Studies

- SUSESO has been testing Machine Learning (ML) algorithms to deal with the high workload involved in processing claims on rejected licences. The model aims to provide the decision making team with a suggested final decision. The project is still being tested before its implementation.
- DNP developed an algorithm to predict missing social information. To improve vulnerability estimations and targeting, they moved from traditional econometric estimations to a ML model.
- Both institutions have decades of sustained work on their internal information systems and data protocols behind them.
- Both SUSESO and DNP needed to hire external experts to train ML models. In Chile, they had to look to the private sector for providers. In Colombia, they sought foreign AI talent from the IDB and MIT.
- Although workforce replacement is a common concern in the context of AI adoption, the mentioned models are designed to manage overwhelming workloads and improve targeting accuracy, rather than reduce team sizes.

Insights from Case Studies (2)

- SUSESO hired a private firm via Chile Compra, the national public procurement agency. As part of this process, SUSESO had to follow the agency's Standard Bidding Terms (Bases Tipo), which stipulated strict standards of transparency, explainability, and bias detection.
- The Chilean case keeps the “human in the loop”. While the predictive model suggest wheter a claim should be accepted or rejected, the specialized team will still be accountable for the decision made.
- The Colombian institution faces a tension between its system's security and transparency. While the model is kept secret to prevent manipulation of information, there is criticism about the algorithm being a black box in a sensitive context such as focalization.
- Because machine learning is highly data intensive, training models on limited or biased datasets risks automating and replicating the exclusion of historically vulnerable groups.
- Whether it is the rejection of a medical claim or a mistaken income presumption for social targeting, the potential for error remains a core concern for global AI ethics within the public sector.